Practical No. 15: Write a program to pass function argument into React component

Conclusion

In this practical, we learned how to pass a function as a prop from a parent component to a child component in React. The child component can invoke the function with arguments, enabling effective communication between components. This concept helps in building dynamic, reusable, and interactive web applications.

Practical Related Questions

✅ **Program 1: Multiple Arguments using Function Prop**

Question:

Write a ReactJS program to demonstrate how multiple arguments can be passed through a function prop from a parent to a child component. Display both arguments in an alert message.

File 1: MultipleArgs.js

```
import React from "react";

// Parent Component
function MultipleArgs() {

  function showMessage(name, age) {
    alert("Name: " + name + ", Age: " + age);
  }

  return <Child sendData={showMessage} />;
}

// Child Component
function Child(props) {
  return (
    <div>
      <h2>Multiple Arguments Example</h2>
      <button onClick={() => props.sendData("Ganesh", 22)}>
        Click Me
      </button>
    </div>
  );
}

export default MultipleArgs;
```

✅ Program 2: Counter Update using Function Prop

Question:

Write a ReactJS program where the child component triggers a parent's function to update a counter value on button click. Display the updated count on the screen.

File 2: Counter.js

```
import React, { useState } from "react";

// Parent Component
function Counter() {

  const [count, setCount] = useState(0);

  function updateCount() {
    setCount(count + 1);
  }

  return (
    <div>
      <h2>Count: {count}</h2>
      <Child increment={updateCount} />
    </div>
  );
}

// Child Component
function Child(props) {
  return (
    <button onClick={props.increment}>
      Increase Count
    </button>
  );
}

export default Counter;
```

✅ **Main File: App.js**

👉 (Run only ONE program at a time using comments)

```
import React from "react";
import MultipleArgs from "./MultipleArgs";
import Counter from "./Counter";

function App() {
  return (
    <div>
      {/* Uncomment ONLY ONE */}

      <MultipleArgs />
      {/* <Counter /> */}

    </div>
  );
}

export default App;
```

XI. Exercise

1. Can we call the same function with different arguments from the child component?

Yes, we can call the same function multiple times with different arguments from the child component. This allows dynamic interaction based on user input.

---

2. How does this concept help in real-world React applications?

- Helps in communication between components
- Improves code reusability
- Makes UI dynamic and interactive
- Useful for handling events (click, form submission)
- Centralizes logic in parent component